Robot as a Peer in Learning
Effect of Robot Role on Help Seeking and Comfort

Likhith Kumar Shivakumar

10-08-2026

1 Introduction

Social robots been showing up in classrooms for a while now, and mostly they get stuck in one of two roles either the tutor, who already knows the answer and just hands it over, or the teacher, who runs the whole lesson from the front of the room. Belpaeme et al. [1] pulled together 101 published studies and 309 individual results on robots in education and found that both roles genuinely helps. Robot tutoring produced a mean cognitive outcome effect size (Cohen's d, weighted by sample size) of 0.70, and a mean affective outcome effect size of 0.59, close to the 0.79 benchmark reported for human one-to-one tutoring, though it doesn't quite match it. A good chunk of that benefit, or so the review argues, comes down to physical presence: people comply more, engage more, and learn faster from an embodied robot compared to the same content delivered by, say, a virtual agent or a video.

What the review also flags, almost in passing really, is a third role that's a lot less explored the robot as a peer. Instead of standing over the learner as some kind of authority figure, a peer robot sits on the same level as the student, figuring things out together with them rather than just dispensing correct answers. The idea being that a peer feels less intimidating than a tutor or a teacher, and peer-to-peer dynamics can carry real advantages over the more evaluative tutor-to-student one. Belpaeme et al. [1] cite early evidence for this: Kanda et al. [3] found children working with a peer-styled robot paid attention longer, responded faster, and answered more accurately than children who were paired with an identical-looking tutor robot, and Zaga et al. [2] report a related peer-versus-tutor comparison of task engagement, also in children.

The catch, though, is this evidence is thin. Across the whole meta-analysis only 9% of the 309 results involved a robot presented as a peer or a novice, the rest split between the tutor role (48%) and the teacher role (38%). Most of what does exist on peer robots was also collected with children, not adults, and the review is explicit that "experimental evidence for this effect remains limited." That gap theoretically promising but empirically thin, especially outside of children's classrooms is the open question this study is trying (on however small a scale) to chip away at.

The open question we picked up from Belpaeme et al. [1], and which frames the whole of this report, is this: does a robot's role as a peer, rather than a tutor, change how often (adult) students ask for help and how comfortable they feel during a learning task? We split this into the two hypotheses that ended up structuring our study:

H1: Students who work with a robot peer will ask for help more often than students who work with a robot tutor.

H2: Students in the peer condition will report higher comfort (and, implicitly, lower stress) than students in the tutor condition even if their task scores turn out similar.

Both hypotheses follow pretty directly from the peer robot rationale Belpaeme et al. sketch out: if a peer framing really does lower the perceived cost of asking for help, we should see it show up both behaviourally (more help requests) and subjectively (higher comfort ratings), without there being a corresponding jump in raw quiz performance, since the mechanism is supposed to be about willingness to ask, not about giving away more of the answer. That last "without a jump in performance" part turned out to be, honestly, the trickiest bit of the whole study, and is really the central finding we want to report honestly rather than paper over.

2 Related Work

Besides the Belpaeme et al. [1] review this whole project is basically anchored to, a handful of more specific studies shaped how we set the quiz up, and it's probably worth being explicit about what we actually borrowed from each one rather than just listing citations one after another.

Zaga, Lohse, Truong and Evers [2] ran one of the studies Belpaeme et al. [1] cite directly for the peer-versus-tutor comparison, looking at how a robot's social character (peer vs. tutor) affects children's task engagement in a guided discovery learning task. Their framing of "social character" as something the robot signals through what it says, rather than through its physical shape or hardware, is basically the design choice we copied our peer and tutor conditions run on the same chat interface and the same quiz content, and the only thing that changes is the wording of the hints and the framing of the welcome message. Had the robot looked different across conditions too, we wouldn't have been able to say whether any effect came from the role, or just from the appearance.

Kanda, Hirano, Eaton and Ishiguro [3] deployed Robovie, one of the first fully autonomous robots placed in an actual elementary school, and found a peer-framed robot produced longer attention and more accurate responses in children than an otherwise identical tutor robot did. This is one of the pieces of evidence Belpaeme et al. [1] lean on when they say peer robots "might" reduce intimidation, and it's more or less where the idea that a peer role could plausibly change accuracy (not just comfort) as a side effect first came from which is more or less exactly what happened in our data too, though for reasons that have more to do with our hint design than any deep social effect (see Section 5.4).

Salomons, Pineda, Adéjàre and Scassellati [4] is the study we leaned on most for the decision to run this with university students rather than children. Their HRI 2022 paper compares a peer robot to a tutor robot teaching adults electronic circuit construction, and finds no significant overall difference in the number of skills learned between conditions, but participants with low prior domain knowledge learned significantly more with the peer robot, and the peer robot was rated friendlier, more social, smarter, and more respectful regardless of prior skill. Two things came out of this for us: first, the peer-versus-tutor question isn't only a children's-classroom question, adults show effects too; second, when peer robots do beat tutor robots on outcomes it tends to be a subgroup or perception effect rather than a blanket one. Which made us cautious about just reading a flat "peer condition scored higher" result at face value, and pushed us toward checking the help-log data in detail rather than trusting the summary statistics alone a decision that ended up mattering quite a bit (Section 5.4).

Diyas, Brakk, Aimambetov and Sandygulova [5] ran a shorter, more applied comparison of a peer versus teacher robot in a programming-learning scenario. We didn't draw heavily on their specific findings, but their design keeping the lesson content identical and varying only the robot's framing is another data point for the same principle Zaga et al. establish, and it's part of why we felt reasonably okay using one shared quiz bank across both our conditions instead of writing two separate quizzes.

Chase, Chin, Oppezzo and Schwartz [6] write about the protégé effect, which is the finding that people invest more effort and learn more when they believe they're teaching or supporting someone rather than being taught themselves. This isn't exactly our design (our robot isn't a novice being taught by the participant), but it's the theoretical backdrop for why a peer robot saying things like "we'll figure this out together" might change a participant's effort and comfort the protégé effect literature suggests even the framing of being a co-learner, rather than being evaluated, can shift how people engage with a task. Our peer hint templates (Section 3.1) were written with this in mind, using "we" and "us" language on purpose instead of "you."

Dillenbourg [7] gives the older, more theoretical grounding for what "collaborative learning" is even supposed to mean, and mostly served as a sanity check on our hint wording: the point that collaboration is defined less by physical co-presence and more by a symmetry of roles and shared goal-orientation is why we tried not always successfully, see Section 5.4 to keep the peer robot's hints looking like joint reasoning ("I'm not sure about this one either") rather than disguised answer-giving.

Bainbridge, Hart, Kim and Scassellati [8] is the paper we go back to in the discussion (Section 6) when talking about the limitation of running this study through a laptop chat window instead of a physically embodied robot: they show physically present robots produce measurably stronger interaction effects than the same robot shown on video, which is one of the central arguments in Belpaeme et al.'s [1] review too, and is a limitation of our own setup since we didn't have equipment to embody the agent physically.

3 System Description

The system is a small Flask web application (`app.py`) serving a single-page quiz interface (`templates/index.html`) that's styled to look like a chat with a robot. Condition assignment happens on the server before the page is even rendered: the `/` route in `app.py` calls `condition = random.choice(['peer', 'tutor'])` and passes the result into the template. The intro screen has no clickable role buttons, instead it just tells the participant directly "Your robot companion has been randomly assigned: Peer (Study Buddy)" or "Tutor (Instructor)." This random, server-side assignment was in place for the whole of data collection, there was no stage, at any point, where a participant picked their own condition. We state this explicitly here because it matters for how Section 4.2 and the results in Section 5 ought to be read: condition membership in the analysed dataset was determined by the server's coin flip, not by participant choice.

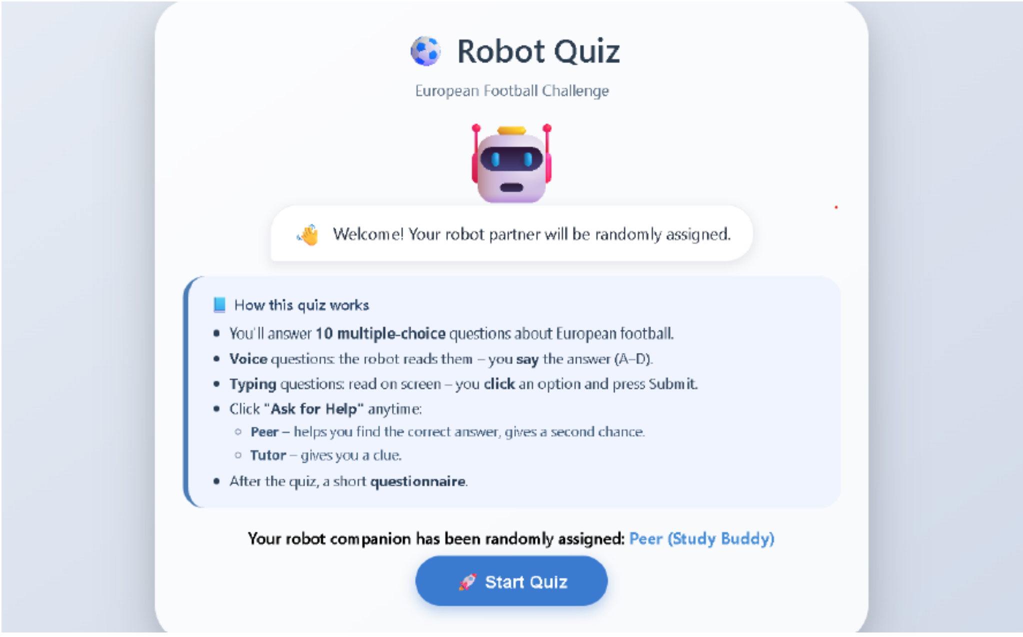

Figure 1: Intro / role assignment screen.

Once the participant clicks "Start Quiz," the robot doesn't jump straight to the first question anymore. It instead delivers a longer, role-specific spoken explanation via the browser's Web Speech API (`speechSynthesis`), and all interaction with the quiz (submitting an answer, asking for help) gets blocked client-side (via an `isExplaining` flag) until that explanation is finished playing. Only once it ends does the quiz area show up and the first question get displayed. This is described fully in Section 3.2, since the content of the two explanations differs enough between conditions that it's worth treating as part of the role manipulation in its own right, not just a cosmetic add-on. After the explanation, each of the ten questions gets presented one at a time with four multiple-choice options; depending on the question index, the interface expects either a spoken answer (via `SpeechRecognition`, with a graceful fallback message if the browser doesn't support it) or a typed/clicked answer.

The quiz content itself is European football trivia questions like which club has won the most Champions League titles, or who was top scorer at the 2022 World Cup split into two ten-item sets (Set A and Set B) handed out to participants at random when the page loads. We picked this domain over something more academic (maths, vocabulary, general science) mainly to keep the barrier to entry low and roughly equal across a mixed group of student participants: everybody either already knows a decent amount of football trivia, or knows basically none, so we weren't too worried about the quiz accidentally testing prior coursework knowledge some participants had and others didn't. It also keeps the mood of the twenty-minute session light, which mattered for a between-subjects design run mostly with people we knew personally a heavier academic quiz risked making the "help me, I'm stuck" dynamic feel like a real exam rather than a game, which would've worked against the comfort measure we were trying to capture.

Question sets A and B are written to be of matched, medium difficulty (roughly one "everybody probably knows this" question, several "known to football fans" questions, and a couple of genuinely obscure ones per set), so any score difference between conditions could be attributed to the robot's role rather than to one group just getting an easier set of questions. Both sets carry a short factual clue field per question (e.g. "They have won it 14 times, more than any other club") which is only ever shown to participants in the tutor condition.

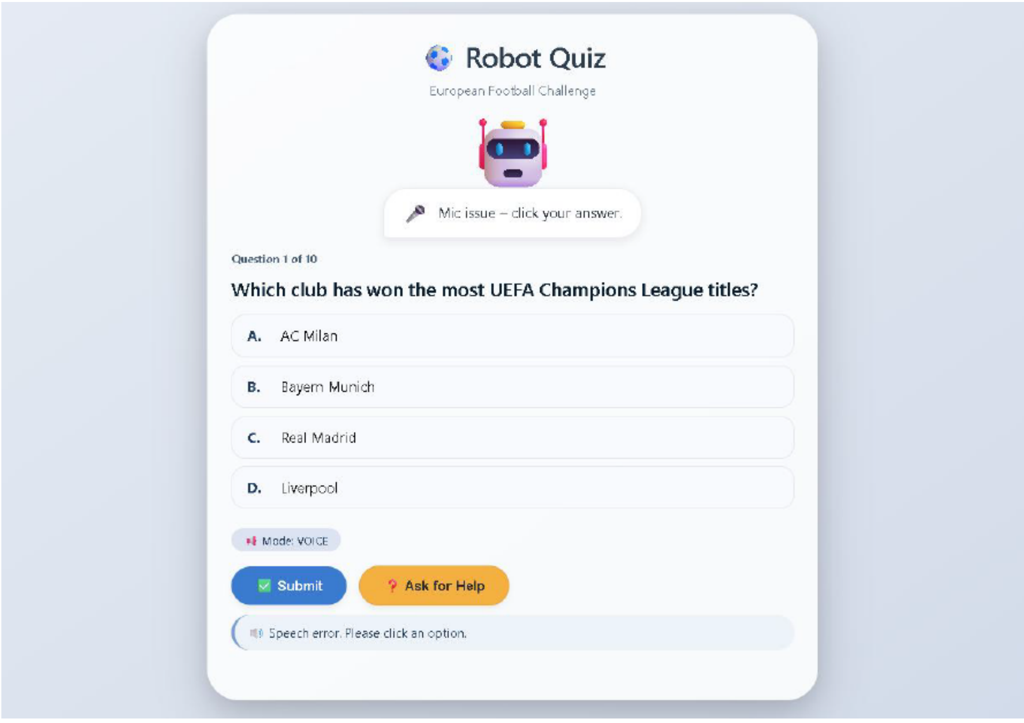

Figure 2: Quiz question screen.

3.1 The Hint Mechanism

This is the part of the system that most directly implements the "peer vs. tutor" manipulation, and as we discuss at length in Section 5.4, is also the part responsible for most of what went wrong with H1 and the control (score) variable. When a participant presses "Ask for Help," the Flask backend (the `/get_hint` route in `app.py`) does two structurally different things depending on the condition:

Peer condition: the server picks one random wrong option out of the remaining answer choices and returns a friendly line telling the participant to rule it out, drawn from a bank of ten templates written in first-person-plural, collaborative language for example: "I've got a feeling {wrong_option} might be a trap. Let's leave it out you're doing great!" The response also tells the frontend which option to visually cross out (`remove_option`), so the participant literally watches one wrong answer disappear off the list. If a participant keeps pressing the button the system keeps eliminating wrong options one at a time, until in the extreme case only the correct option is left standing.

Tutor condition: the server doesn't touch the answer options at all. It returns one of four short lines that surface the question's pre-written factual clue, for example: "Here's a clue: {clue}" which requires the participant to still reason from a hint toward an answer, rather than just being handed a shrinking set of options.

On top of that asymmetry, the peer condition also gives a second attempt after a first wrong answer before the correct answer gets revealed (a small retry allowance, implemented client-side as `retryCount` in `index.html`), while the tutor condition gives immediate right/wrong feedback with no retry at all. Both of these design choices were made with good intentions meant to make the peer robot feel more forgiving and supportive, in keeping with the "we'll figure this out together" framing but in hindsight they also mean the peer condition is, mechanically, just an easier version of the same quiz, independent of anything to do with how "peer-like" the robot feels. We flag this here, at the system-description stage, because it's essential context for reading the results honestly, not something we want to bury in a limitations paragraph right at the end.

3.2 The Pre-Quiz Role Explanation

Both conditions now hear a scripted spoken explanation immediately after clicking "Start Quiz," before the first question even appears. The intent, same as with the welcome line it replaces, was to extend the peer/tutor manipulation all the way to the very opening of the session rather than confining it to the hint system alone something closer to how a real tutor or a real study partner would set the scene before getting started, instead of just a bare instruction screen.

In practice though, the two scripted explanations aren't matched in length or content, and this is worth documenting carefully rather than glossing over, since it's a new source of asymmetry between conditions that sits right alongside the ones already described in Section 3.1.

Figure 3: The peer condition's pre-quiz explanation screen, shown (and read aloud via speechSynthesis) immediately after "Start Quiz" is clicked.

The tutor explanation runs to roughly 75 words and is written in a formal, instructional register ("Good day, student. Today we will explore the fascinating history of European football... I expect you to think carefully"), framing the session as a graded exercise and mentioning, once, that a clue will be available on request. The peer explanation runs to roughly 110 words about 45% longer and is written in a casual, enthusiastic register ("Hey! So you're into football? Me too!… we'll figure it out together"). The screenshot below shows exactly what a participant in the peer condition actually saw and heard read out loud at this point in the session.

Beyond just tone, the peer explanation also does something the tutor explanation never does: it explicitly previews the retry mechanic before the quiz has even started ("if you get it wrong the first time, you get another chance to try again"). Meaning peer-condition participants go into the quiz already knowing they've got a safety net that tutor-condition participants are never told about in advance, which plausibly affects how relaxed they feel walking in, independent of anything about the robot feeling like a peer versus an authority figure. We treat this as an additional limitation in Section 6.1 rather than folding it silently into the comfort result.

3.3 An Abandoned LLM-Based Hint Generator

Before settling on the fixed-template system described above we tried, briefly, generating the peer robot's hints dynamically using a small local language model (DistilGPT-2, via the Hugging Face `transformers` pipeline), with the idea being to get more varied, more conversational-sounding hints instead of just ten fixed strings. This did not, in practice, work well enough to keep: response latency sat around three to five seconds per hint on the hardware we had, which was long enough that participants noticed the delay and actually commented on it; the generated text was frequently unreliable too, occasionally inventing facts or flat-out contradicting the correct answer, which obviously isn't acceptable for something meant to help a participant reason toward a correct answer; and running the model locally used enough CPU and memory that it strained the laptops we were using to host sessions. We also realised non-deterministic, free-text hints would've made the `/log_help` data a lot harder to analyse consistently, since we'd have had to code hint content by hand instead of just logging which fixed template got used.

The fallback we ended up shipping a bank of pre-written templates chosen at random, with placeholders filled in from the question data isn't a particularly novel idea, but it solved the practical problems: response time dropped to well under a hundred milliseconds, hints stayed factually correct by construction (we wrote them ourselves, in advance), and every hint given could be logged and traced back to a specific template, which made it possible to reconstruct exactly what happened during each participant's help-seeking behaviour after the fact (see Section 5.4, where this logging is what let us notice the option-elimination gaming pattern in the first place).

4 Methods

4.1 Participants

Fourteen participants took part ($n - 14$), seven assigned to the peer condition and seven to the tutor condition. Ages ranged from 23 to 26 years (all university-affiliated adults); the sample skewed heavily male (12 of 14 participants), which we note explicitly as a demographic limitation rather than glossing over it with only two female participants in the whole dataset we can't really say anything meaningful about gender as a factor here.

Worth being upfront too that this sample is smaller than the $n - 20$–24 range (10–12 per condition) specified in the accompanying ethics application draft, though it is in fact larger than the $n - 6$ (3 per condition) pilot sample described in the original study concept slides.

4.2 Design and Condition Assignment

The study used a between-subjects design with one independent variable, robot role, at two levels (peer, tutor), to avoid the carry-over effects a within-subjects design would've introduced if the same participant experienced both a friendly and an authority-styled robot back to back.

Condition was assigned on the server, before the page was rendered (`condition - random.choice('peer', 'tutor')`), and the intro screen simply told the participant which role they'd been assigned rather than letting them pick. This matches what the ethics application draft and the original slide deck describe ("random assignment to one condition (peer or tutor) by coin flip"), and it's the procedure used to collect the full $n - 14$ dataset analysed in Section 5: every participant's condition was determined by the server's random choice, with no self-selection step at any stage of data collection. The hint-mechanism and retry-count asymmetries described in Section 3.1, and the explanation-length asymmetry described in Section 3.2, remain present in the current code and are the confounds that actually need addressing (Section 6), rather than condition assignment itself.

4.3 Materials and Procedure

Materials consisted of the chat-style web interface described in Section 3, the ten-question quiz (Set A or Set B, assigned at random per session), and a short post-task questionnaire covering a comfort rating (1 to 7 scale), one manipulation-check item, and two demographic questions (age, gender). The procedure for each session went as follows: participants read a short information sheet and, in sessions run under the ethics protocol, signed a consent form; they were then randomly assigned a condition by the server, received the condition-specific spoken welcome message, and were told they could ask the robot for help at any point during the quiz. They then worked through the ten questions one at a time, with help requests, retries, and the final answer to every question logged automatically to `experiment_data.csv` and `help_log.csv`. After the tenth question, participants completed the post-task questionnaire and were thanked. Full sessions ran to roughly 20 minutes as planned. Sessions were conducted remotely, with participants accessing the quiz interface over a locally hosted server made available to them for the duration of their session.

4.4 Measures

One small terminology note: the ethics application draft describes the task-score variable as "percentage correct," but the system as implemented logs and analyses a raw count out of ten instead. Doesn't change the statistics (it's a linear rescaling), but worth mentioning for consistency between the planning documents and what actually got measured.

Table 1: Dependent variables and measures

Variable	Measurement	Purpose
DV1 Help requests	Count of "Ask for Help" presses during the quiz, logged automatically per question	Main test for H1
DV2 Comfort rating	1 7 scale, post-task questionnaire	Main test for H2
DV3 Task score	Number of questions answered correctly (0 10)	Control check
Manipulation check	"Did the robot act more like a classmate or more like a teacher?"	Confirms role was perceived as intended

4.5 Analysis Plan

All three dependent variables were compared between conditions using independent-samples t-tests (`scipy.stats.ttest_ind`), with a significance threshold of $\alpha - .05$. Following the pre-registered, directional hypotheses, H1 (help requests) and H2 (comfort) were tested one-tailed (peer > tutor); the control variable (score) was tested two-tailed, since no directional prediction was made for it in fact the prediction, following Belpaeme et al.'s [1] framing of the peer role, was that scores should come out similar across conditions. Effect sizes were computed as Cohen's d for all three comparisons. We additionally went back and re-examined the manipulation-check responses and the raw `help_log.csv` entries, which is what surfaced the option-elimination pattern discussed in Section 5.4 something a t-test on the aggregate help count alone wouldn't have revealed.

5 Results

Table 2 gives the descriptive statistics for all three dependent variables by condition ($n-7$ per group).

Table 2: Descriptive statistics by condition

Condition	n	Score (M ± SD)	Help Count (M ± SD)	Comfort (M ± SD)
Peer	7	7.29 ± 2.36	12.14 ± 11.84	5.71 ± 0.95
Tutor	7	3.57 ± 1.72	6.00 ± 3.21	3.57 ± 0.79

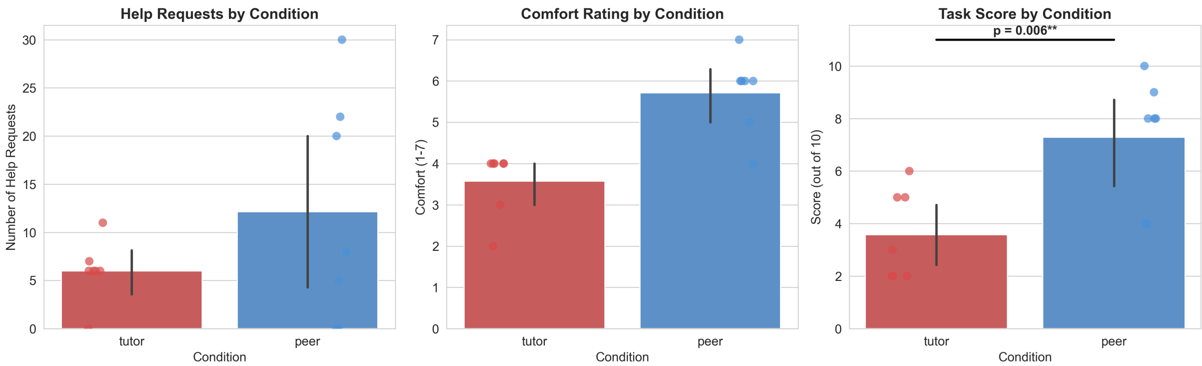

Figure 4: Help requests, comfort ratings, and task scores by condition.

5.1 H1 Help-Seeking Behaviour

Peer-condition participants requested help more often on average ($M - 12.14$, $SD - 11.84$) than tutor-condition participants did ($M - 6.00$, $SD - 3.21$), a descriptive difference of roughly a factor of two. The one-tailed independent-samples t-test on this comparison was not statistically significant, $t(12) - 1.325$, $p - .105$, though the effect size landed in the medium-to-large range ($d - 0.71$). H1 is therefore not supported, at least not at the conventional $\alpha - .05$ threshold.

The reason it falls short of significance despite the fairly large mean difference is visible directly in Figure 4 (left panel): the peer group's help-count data is extremely spread out, ranging from 0 requests (two participants who apparently never asked for help at all) up to 30 requests from a single participant, whereas the tutor group is much more tightly clustered (0 to 11 requests). With only seven participants per group, one high outlier is enough to inflate both the mean and the standard deviation substantially, which is exactly what happened here, and is part of why we went back to `help_log.csv` directly instead of just trusting the summary numbers (see Section 5.4).

5.2 H2 Comfort

Peer-condition participants reported markedly higher comfort ($M - 5.71$, $SD - 0.95$, on the 1-to-7 scale) than tutor-condition participants ($M - 3.57$, $SD - 0.79$). This difference was statistically significant, $t(12) - 4.593$, $p < .001$, one-tailed, with a very large effect size ($d - 2.46$). Taken at face value this is the cleanest result in the study, and the one that most directly supports the peer-robot rationale from Belpaeme et al. [1]: participants who worked with the "study buddy" framing rated the experience as noticeably more comfortable than those who worked with the "instructor" framing. We come back to why we don't think this can be cleanly attributed to the peer framing alone in Section 5.4 and Section 6.

5.3 Control Variable Task Score

This is where the study runs into its main problem, really. The control variable was expected, following the logic of H2 ("even if task scores are similar"), to show no meaningful difference between conditions the peer framing was supposed to change how participants felt about asking for help, not how many questions they actually got right. Instead, peer-condition participants scored substantially and significantly higher ($M - 7.29$ out of 10, $SD - 2.36$) than tutor-condition participants ($M - 3.57$, $SD - 1.72$), $t(12) - 3.366$, $p - .006$, two-tailed, $d - 1.80$ (also annotated directly on Figure 4, right panel).

This is a large effect on a variable that was supposed to be the study's null result, and it directly undermines a clean reading of the comfort finding in Section 5.2, because success and comfort are entangled here: doing almost twice as well on the quiz is, on its own, a perfectly good explanation for feeling more comfortable afterward, independent of anything about how "peer-like" the interaction felt. Section 5.4 explains why we think this score gap is close to a mechanical artefact of how the two hint systems were built (Section 3.1), rather than a genuine cognitive or social effect of the robot's role.

5.4 Why the Score and Help-Count Results Look the Way They Do

Going back to the raw `help_log.csv` entries makes the score result a lot easier to explain, and also complicates the H1 story somewhat. One peer-condition participant (participant 2007) requested help exactly three times on almost every question the maximum useful number of requests, since three of the four options can be eliminated one at a time before only the correct answer is left standing. The logged hint text for this participant follows the same pattern across nearly every question, for instance eliminating three different wrong options on question 1 in three consecutive requests a few seconds apart. This participant went on to score 10 out of 10 with 30 logged help requests the single largest score, and by far the largest help count, in the whole dataset, and a clear driver of both the peer group's inflated mean help count (Section 5.1) and part of its inflated mean score (Section 5.3).

This isn't really "help-seeking" in the sense the study set out to measure, if we're being honest about it. Because the peer condition's hint mechanism directly removes a wrong option rather than giving an indirect clue, and because it doesn't limit how many times a participant can ask, the system effectively lets a participant brute-force the correct answer by process of elimination, three help requests at a time. The tutor condition's hint mechanism, by contrast, only ever surfaces the same short textual clue regardless of how many times it's requested, so repeated requests there can't mechanically raise a participant's odds of getting the answer right the way they can in the peer condition. Once this gets taken into account, the higher average score in the peer condition looks a lot less like evidence that peer framing improves learning outcomes, and a lot more like a direct, mechanical consequence of how much more informative each peer help request is compared to each tutor help request combined with the peer condition's extra retry attempt on a first wrong answer (Section 3.1), which the tutor condition never gets either.

For H1 specifically, this also means the raw help-count comparison is measuring two different things folded together: participants who asked for help because they were genuinely unsure and wanted support (the behaviour H1 was designed to capture), and at least one participant who appears to have been using the help button strategically, as a way to solve the quiz rather than to get reassurance. The original study materials for this project explicitly note this exact pattern ("Gaming Behaviour: Peer participants using help systematically to eliminate options") as something discovered from the help log so it's not a new observation on our part, but it's one that, in our view, deserved more attention in the write-up than a single bullet point, since it goes a long way toward explaining why H1 came out non-significant despite a large raw mean difference .

5.5 Manipulation Check

All fourteen participants answered the manipulation-check question ("Did the robot act more like a classmate or more like a teacher?") in a way that matched their assigned condition: all seven peer-condition participants described the robot as a classmate, all seven tutor-condition participants described it as a teacher. No participants were excluded on this basis.

6 Discussion and Conclusion

Coming back to the research question from Section 1 does a robot's role as a peer, rather than a tutor, change how often students ask for help and how comfortable they feel during a learning task our answer has to be a genuinely mixed one, and we think it's more useful to just say that plainly than to round it up to a clean "yes." H2 (comfort) is supported by the numbers: peer-condition participants rated the experience as significantly more comfortable, with a large effect size. H1 (help-seeking) is not supported at the conventional significance threshold, despite peer-condition participants asking for help roughly twice as often on average, because the peer group's help counts were extremely variable and driven substantially by one participant's strategic, option-elimination use of the help button rather than by a broad, condition-wide shift in willingness to ask for support.

More importantly, we don't think the comfort result can be attributed cleanly to "feeling like the robot is a peer rather than an authority figure," which is what H2 was actually trying to test. Because the peer condition's hint system gives away more information per request (eliminating a wrong option outright, versus a textual clue that still has to be reasoned about) and grants an extra retry attempt the tutor condition doesn't get, peer-condition participants ended up scoring almost twice as high on the quiz a result the study design explicitly did not predict, and one that, per the original hypothesis wording ("even if task scores are similar"), was specifically supposed to be ruled out. Since doing well on a quiz is, on its own, a reasonable explanation for feeling comfortable about the experience afterward, our comfort result is confounded with our score result, and we cannot cleanly separate "participants felt more comfortable because the robot felt like a peer" from "participants felt more comfortable because the peer condition's hint system made the quiz substantially easier to do well on."

This pattern is also worth setting against Salomons et al. [4], whose comparison of peer and tutor robots teaching adults found no significant overall difference in learning outcomes between conditions, with only a subgroup (low prior domain knowledge) benefiting from the peer robot. Our finding of a large, condition-wide score gap sits a bit awkwardly next to that result, and together with the option-elimination pattern documented directly in our own help log, it points toward the score gap being a property of our particular hint implementation rather than a replication of a genuine social effect of peer framing on learning.

6.1 Limitations

- The two hint mechanisms are not informationally equivalent. The peer hint removes a wrong answer option outright and can be requested repeatedly without limit; the tutor hint only ever surfaces a fixed textual clue. This is very likely the main driver of the significant score difference in Section 5.3, and by extension complicates the comfort result in Section 5.2.

- The pre-quiz spoken role explanation (Section 3.2) is not matched between conditions: the peer explanation is roughly 45% longer and explicitly previews the retry mechanic before the quiz begins, while the tutor explanation doesn't mention any equivalent allowance. Any future study should either equalise the length and informational content of the two explanations, or remove the retry preview, since as written it could shape expectations (and therefore comfort) independently of the peer/tutor framing itself.

- The peer condition includes a second-attempt allowance after a first wrong answer; the tutor condition does not. This is a further, independent source of the score gap between conditions.

- Sample size ($n - 14$, 7 per condition) is small, sits below the $n - 20$–24 range specified in the ethics application draft, and is likely underpowered for detecting a moderate effect on the noisy help-count variable in particular, as the non-significant but medium-to-large effect size for H1 suggests.

- The interaction was delivered through a laptop chat window rather than a physically embodied robot; Belpaeme et al. [1] and Bainbridge et al. [8] both report that physical embodiment strengthens social robot effects, so our results may understate, or otherwise differ from, what a fully embodied peer robot would produce.

6.2 Future Work

A cleaner replication of this study would equalise the information content of the two hint conditions (for example, giving both a single textual clue, or both a single option-elimination, rather than mixing the two mechanisms across conditions), remove the retry-attempt asymmetry, equalise the length and content of the pre-quiz role explanations described in Section 3.2, and collect the full planned sample of 20–24 participants with a more balanced gender mix. Condition assignment itself was already randomised server-side throughout data collection and isn't really a concern for future replications; it's the hint, retry, and explanation asymmetries above that would actually need addressing. It would also be worth adding a rate limit, or some explicit "cost," to repeated help requests, so the help-count variable measures something closer to felt need for support rather than an available strategy for just solving the quiz. Given the resources, running the study with a physically embodied robot rather than a browser-based chat agent, following the embodiment argument in Belpaeme et al. [1] and Bainbridge et al. [8], would also be a natural next step.

Taken together, we think this pilot adds a small, adult, and admittedly messy data point to the gap Belpaeme et al. [1] identify only 9% of the studies in their review involve a robot presented as a peer at all. What we'd flag most strongly for anyone building on this work, ourselves included, is less about the specific numbers and more about the design trap we fell into: it's very easy, when trying to make a "peer" hint feel supportive and forgiving, to accidentally also make it more informative than the "tutor" hint it's being compared against at which point any resulting difference in comfort or performance stops being evidence about the social framing, and starts being evidence about who just got handed more of the answer.

References

[1] T. Belpaeme, J. Kennedy, A. Ramachandran, B. Scassellati, and F. Tanaka, "Social robots for education: A review," *Sci. Robot.*, vol. 3, no. 21, eaat5954, Aug. 2018.

[2] C. Zaga, M. Lohse, K. P. Truong, and V. Evers, "The effect of a robot's social character on children's task engagement: Peer versus tutor," in *Proc. Int. Conf. Social Robotics (ICSR)*, Springer, 2015, pp. 704 713.

[3] T. Kanda, T. Hirano, D. Eaton, and H. Ishiguro, "Interactive robots as social partners and peer tutors for children: A field trial," *Human Computer Interaction*, vol. 19, pp. 61 84, 2004.

[4] N. Salomons, K. T. Pineda, A. Adéjàre, and B. Scassellati, " 'We Make a Great Team!': Adults with Low Prior Domain Knowledge Learn more from a Peer Robot than a Tutor Robot," in *Proc. ACM/IEEE Int. Conf. Human Robot Interaction (HRI)*, 2022.

[5] Y. Diyas, D. Brakk, Y. Aimambetov, and A. Sandygulova, "Evaluating peer versus teacher robot within educational scenario of programming learning," in *Proc. 11th ACM/IEEE Int. Conf. Human Robot Interaction (HRI)*, 2016, pp. 425 426.

[6] C. C. Chase, D. B. Chin, M. A. Oppezzo, and D. L. Schwartz, "Teachable agents and the protégé effect: Increasing the effort towards learning," *J. Sci. Educ. Technol.*, vol. 18, pp. 334 352, 2009.

[7] P. Dillenbourg, "What do you mean by 'collaborative learning'?" in *Collaborative Learning: Cognitive and Computational Approaches*, Pergamon, 1999, pp. 1 19.

[8] W. A. Bainbridge, J. W. Hart, E. S. Kim, and B. Scassellati, "The benefits of interactions with physically present robots over video-displayed agents," *Int. J. Soc. Robot.*, vol. 3, pp. 41 52, 2011.

Proof. **AI declaration**

I used Claude and DeepSeek to help with this report and study setup, mainly to smooth out my writing, fix grammar, and handle LaTeX formatting. The research design, data collection, analysis, results, and conclusions are all by own. Claude didn't contribute any content, data, or analysis, just language and formatting help. Also helped me with researching the study. I used Grammarly too because it help me in framing sentences for faster typing.

□